# CHECKING UNIFORM CATEGORICAL MODULI SPACES ON AFFINES

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 3276-3283 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 3285-3309, the complete author proof (23 body lines).

*Editorial note.* Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 38 of queue_stacks-stacks-more-morphisms.json (pb=1161, stmt 3276-3283, proof 3285-3309).

*Editorial note.* stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 0261: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3309–3309. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

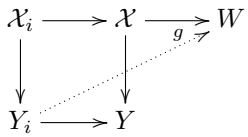
**Lemma 0.1.** *Let $f : \mathcal{X} \to Y$ be a morphism from an algebraic stack to an algebraic space. If for every affine scheme $Y'$ and flat morphism $Y' \to Y$ the base change $f' : Y' \times_Y \mathcal{X} \to Y'$ is a categorical moduli space, then $f$ is a uniform categorical moduli space.*

*Proof.* Choose an étale covering $\{Y_i \to Y\}$ where $Y_i$ is an affine scheme. For each $i$ and $j$ choose a affine open covering $Y_i \times_Y Y_j = \bigcup Y_{ijk}$. Set $\mathcal{X}_i = Y_i \times_Y \mathcal{X}$ and $\mathcal{X}_{ijk} = Y_{ijk} \times_Y \mathcal{X}$. Let $g : \mathcal{X} \to W$ be a morphism towards an algebraic space.

---

*Date*: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Then we consider the diagram




The assumption that $\mathcal{X}_i \to Y_i$ is a categorical moduli space, produces a unique dotted arrow $h_i : Y_i \to W$. The assumption that $\mathcal{X}_{ijk} \to Y_{ijk}$ is a categorical moduli space, implies the restriction of $h_i$ and $h_j$ to $Y_{ijk}$ are equal. Hence $h_i$ and $h_j$ agree on $Y_i \times_Y Y_j$. Since $Y = \coprod Y_i / \coprod Y_i \times_Y Y_j$ (by Spaces, Section 0261) we conclude that there is a unique morphism $Y \to W$ through which $g$ factors. Thus $f$ is a categorical moduli space. The same argument applies after a flat base change, hence $f$ is a uniform categorical moduli space. $\square$